YUWEI ZHANG

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EDUCATION

University of California, San Diego, San Diego, CA, U.S.

Ph.D. in Data Science, Advisor: Prof. Jingbo Shang *Sept, 2023 – 2027*

University of California, San Diego, San Diego, CA, U.S.

M.S. in Electrical Engineering, track: Machine Learning & Data Science *Sept, 2021 – Jun, 2023*

- **GPA: 4.0/4.0**
- **Selected Courses:** Programming for Data Analysis (A+), Advanced Statistical NLP (A+), GPU Programming (A), Recommender Sys&Web Mining (A+), Probabilistic Reason&Learning (A)

Nankai University, Tianjin, P.R.China

B.S. in Physics *Aug, 2016 – Jun, 2020*

- Graduate from **Physics Boling Class**, Nankai University

SELECTED PUBLICATIONS

* Equal contribution.

- Xintong Li, Sha Li, **Yuwei Zhang**, Changlong Yu, Rongmei Lin, Hongye Jin, Shuyi Guan, Xin Liu, Linwei Li, Qingyu Yin, Jingbo Shang. Process Reward Informed Tree Rollout for Effective Multi-Turn RL. *Arxiv 2026*. [paper]
- Haoran Liu*, **Yuwei Zhang***, Xiyao Li, Bohan Lyu, Jingbo Shang. HERO: Hindsight-Enhanced Reflection from Environment Observations for Agentic Self-Distillation. *Arxiv 2026*. [paper] [code]
- **Yuwei Zhang**, Sha Li, Changlong Yu, Qin Lu, Shuowei Jin, Chengyu Dong, Haoran Liu, Ilgee Hong, Xintong Li, Zhenyu Shi, Bing Yin, Jingbo Shang. Learning with Rare Success but Rich Feedback via Reflection-Enhanced Self-Distillation. *Arxiv 2026*. [paper] [blog post] [code]
- Xintong Li, Sha Li, Rongmei Lin, Hongye Jin, Linwei Li, Hejie Cui, Sarah Zhang, Chia-Yuan Chang, Kewei Cheng, **Yuwei Zhang**, Qingyu Yin, Jingbo Shang, Bing Yin. Stepwise Penalization for Length-Efficient Chain-of-Thought Reasoning. *NeurIPS 2026*. [paper]
- Guangchen Lan, Sipeng Zhang, Tianle Wang, **Yuwei Zhang**, Daoan Zhang, Xinpeng Wei, Xiaoman Pan, Hongming Zhang, Dong-Jun Han, Christopher G. Brinton. MAPPO: Maximum a Posteriori Preference Optimization with Prior Knowledge. *NeurIPS 2026*. [paper]
- **Yuwei Zhang**, Chengyu Dong, Shuowei Jin, Changlong Yu, Hejie Cui, Hongye Jin, Xinyang Zhang, Hamed Bonab, Colin Lockard, Jianshu Chen, Zhenyu Shi, Jingbo Shang, Xian Li, Bing Yin. CoMem: Context Management with A Decoupled Long-Context Model. *ICML 2026*. [paper] [code]
- **Yuwei Zhang**, Wenhao Yu, Shangbin Feng, Yifan Zhu, Letian Peng, Jayanth Srinivasa, Gaowen Liu, Jingbo Shang. Bidirectional LMs are Better Knowledge Memorizers? A Benchmark for Real-World Knowledge Injection. *ACL 2026*. [paper] [code]
- Zhongkai Yu, Yichen Lin, Chenyang Zhou, **Yuwei Zhang**, Kun Zhou, Junxia Cui, Haotian Ye, Zhengding Hu, Zaifeng Pan, Ruiyi Wang, Yujie Zhao, Hejia Zhang, Jingbo Shang, Jishen Zhao, Yufei Ding. ChipMATE: Multi-Agent Training via Reinforcement Learning for Enhanced RTL Generation. *Arxiv 2026*. [paper] [code]

- Xintong Li, Jalend Bantupalli, Ria Dharmani, **Yuwei Zhang**, Jingbo Shang. Toward Multi-Session Personalized Conversation: A Large-Scale Dataset and Hierarchical Tree Framework for Implicit Reasoning. *EMNLP 2025*. [paper]
- **Yuwei Zhang**, Jayanth Srinivasa, Gaowen Liu, Jingbo Shang. Attention Reveals More Than Tokens: Training-free Long-context Reasoning with Attention-guided Retrieval. *Arxiv 2025*. [paper]
- Di Wu, Hongwei Wang, Wenhao Yu, **Yuwei Zhang**, Kai-Wei Chang, Dong Yu. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. *ICLR 2025*. [paper]
- Zilong Wang, Zifeng Wang, Long Le, Huaixiu Steven Zheng, Swaroop Mishra, Vincent Perot, **Yuwei Zhang**, Anush Mattapalli, Ankur Taly, Jingbo Shang, Chen-Yu Lee, Tomas Pfister. Speculative RAG: Enhancing Retrieval Augmented Generation through Drafting. *ICLR 2025*. [paper]
- Letian Peng*, **Yuwei Zhang***, Zilong Wang, Jayanth Srinivasa, Gaowen Liu, Zihan Wang, Jingbo Shang. Answer is All You Need: Instruction-following Text Embedding via Answering the Question. *ACL 2024*. [paper] [code]
- **Yuwei Zhang**, Siffi Singh, Sailik Sengupta, Igor Shalyminov, Hang Su, Hwanjun Song, Saab Mansour. Can Your Model Tell a Negation from an Implicature? Unravelling Challenges With Intent Encoders. *ACL 2024*. [paper] [code]
- Letian Peng, **Yuwei Zhang**, Jingbo Shang. Controllable Data Augmentation for Few-Shot Text Mining with Chain-of-Thought Attribute Manipulation. *Findings of ACL 2024*. [paper] [code]
- **Yuwei Zhang**, Zihan Wang, Jingbo Shang. ClusterLLM: Large Language Models as a Guide for Text Clustering. *EMNLP 2023*. [paper] [code]
- **Yuwei Zhang***, Chih-Hui Ho*, Nuno Vasconcelos. Toward Unsupervised Realistic Visual Question Answering. *ICCV 2023*. [paper] [code]
- Haode Zhang, Haowen Liang, **Yuwei Zhang**, Liming Zhan, Xiao-Ming Wu, Xiaolei Lu, Albert Y.S. Lam. Fine-tuning Pre-trained Language Models for Few-shot Intent Detection: Supervised Pre-training and Isotropization. *NAACL 2022*. [paper] [code]
- **Yuwei Zhang**, Haode Zhang, Li-Ming Zhan, Xiao-Ming Wu, Albert Y.S. Lam. New Intent Discovery with Pre-training and Contrastive Learning. *ACL 2022*. [paper] [code]
- Haode Zhang*, **Yuwei Zhang***, Li-Ming Zhan, Jiaxin Chen, Guangyuan Shi, Xiao-Ming Wu, Albert Y.S. Lam. Effectiveness of Pre-training for Few-shot Intent Classification. *Findings of EMNLP 2021 (short)*. [paper] [code]
- **Yuwei Zhang***, Shuai Xu*, Ronghua Zhang, Zhichao Deng, Yin Liu, Jianguo Tian, Li Yu, Qiongzheng Hu, Qing Ye. Automated Calculation of Liquid Crystal Sensing Images Based on Deep Learning. *Analytical Chemistry*. [paper]
- Jing Li*, **Yuwei Zhang***, Qing Chen, Zhenhua Pan, Jun Chen, Meixiu Sun, Junfeng Wang, Yingxin Li, Qing Ye. Development and Validation of a Screening Model for Lung Cancer Using Machine Learning: A Large-Scale, Multi-Center Study of Biomarkers in Breath. *Frontiers in Oncology*. [paper]
- Chengjin Wang*, **Yuwei Zhang***, Shuai Xu, Yuyan Liu, Lindan Xie, Changlong Wu, Qianhui Yang, Yanhua Chu, Qing Ye. Research on Assistant Diagnosis of Fundus Optic Neuropathy Based on Deep Learning. *Current Eye Research*. [paper]
- Matthew Ricci, Minju Jung, **Yuwei Zhang**, Mathieu Chalvidal, Aneri Soni, Thomas Serre. KuraNet: Systems of Coupled Oscillators that Learn to Synchronize. *Arxiv 2021*. [paper] [code]

RESEARCH EXPERIENCE

Google Deepmind, CA, U.S. *summer 2026*
Job Title: Student Researcher
Project: Agent Post-Training

Amazon Rufus (Foundation Models Team), CA, U.S. *summer 2025 - spring 2026*
Job Title: Applied Scientist Intern
Project: Agent Post-Training and Context Management

AMD Research, CA, U.S. *spring 2025*
Job Title: Research Scientist Intern
Project: Reinforcement Learning for VLMs

Tencent America, WA, U.S. *summer 2024*
Job Title: Research Scientist Intern
Project: Long-Term Memory and Knowledge Injection

Amazon AWS, WA, U.S. *summer 2023*
Job Title: Applied Scientist Intern
Project: Representation Learning

University of California, San Diego, CA, U.S.
Shang Lab, Department of Computer Science and Engineering *Dec. 2022 – Mar. 2023*
Research Assistant, Advisor: Prof. Jingbo Shang
Project: Representation Learning

University of California, San Diego, CA, U.S.
SVCL Lab, Department of Electrical and Computer Engineering *Jan. 2022 – Nov. 2022*
Research Assistant, Advisor: Prof. Nuno Vasconcelos
Project: Unanswerable Visual Question Answering

The Hong Kong Polytechnic University, HK S.A.R
Department of Computing *Apr. 2021 – Aug. 2021*
Research Assistant, Advisor: Prof. Xiaoming Wu
Project: Representation Learning

Nankai University, Tianjin, P.R. China
School of Physics *Jun. 2020 – Nov. 2020*
Research Assistant, Advisor: Prof. Qing Ye
Project: AI for Physics

Brown University, RI, U.S.
Serre Lab, Carney Institute for Brain Science *Aug. 2019 – Dec. 2019*
Research Assistant, Advisor: Prof. Thomas Serre
Project: AI for Neuroscience

TEACHING

TA Fall 2026 DSC 240 Machine Learning

TA Winter 2025 DSC 148 Introduction to Data Mining

TA Spring 2023 DSC 258R Natural Language Processing

REVIEWER

ACL ARR, ICLR, CVPR, ICML, NeurIPS